9 **(a)** Show that if $a = 1$ and $b > 1$ then $a^b < b^a$. **[2]**

(b) Find integer values of a and b with $b > a > 1$ and a^b not greater than b^a (a counter example to the conjecture given in lines 7–8). **[1]**

9(a)

9(b)